VLANS & STATIC ROUTING

1. In Cisco Packet Tracer, create a simple two-switch network with three VLANs (e.g., VLAN 10 for 'Foodies', VLAN 20 for 'Gamers', VLAN 30 for 'Shoppers'), assign three PCs to each VLAN, and verify that PCs in the same VLAN can ping each other but not PCs in other VLANs.

Answer & Configuration:

To set up the network, deploy two Cisco 2960 switches and 9 PCs in Cisco Packet Tracer. Configure VLANs on both switches:

```
Switch> enable
Switch# configure terminal
Switch(config)# vlan 10
Switch(config-vlan)# name Foodies
Switch(config-vlan)# vlan 20
Switch(config-vlan)# name Gamers
Switch(config-vlan)# vlan 30
Switch(config-vlan)# name Shoppers
Switch(config-vlan)# exit

! Assign ports to VLAN 10, 20, 30 on Switch 1
Switch(config)# interface range fastEthernet 0/1 - 3
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config)# interface range fastEthernet 0/4 - 6
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config)# interface range fastEthernet 0/7 - 9
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
```

Verification: Assign IP addresses in the same subnet (e.g., 192.168.10.x/24 for VLAN 10). Pinging between PCs in VLAN 10 succeeds, but pinging between VLAN 10 and VLAN 20 fails due to Layer 2 isolation.

2. Configure trunk ports between the two switches in your Packet Tracer network so that allVLANs can communicate across both switches.

Answer & Configuration:

Connect the two switches using port FastEthernet 0/24 on both sides. Configure them as 802.1Q trunk ports:

```
Switch(config)# interface fastEthernet 0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk allowed vlan 10,20,30
Switch(config-if)# no shutdown
```

Verification: Run `show interfaces trunk` to confirm trunking is active. PCs in the same VLAN on different switches can now communicate successfully.

3. Add a router to your Packet Tracer network and configure Inter-VLAN routing so that a PCin VLAN 10 can successfully ping a PC in VLAN 20.

Answer & Configuration:

Connect a router (e.g., Cisco 1941) to Switch 1 using a Router-on-a-Stick setup on interface GigabitEthernet 0/0:

```
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# no shutdown

Router(config)# interface gigabitEthernet 0/0.10
Router(config-subif)# encapsulation dot1Q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0

Router(config)# interface gigabitEthernet 0/0.20
Router(config-subif)# encapsulation dot1Q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0

Router(config)# interface gigabitEthernet 0/0.30
Router(config-subif)# encapsulation dot1Q 30
Router(config-subif)# ip address 192.168.30.1 255.255.255.0
```

Configure the switch port connecting to the router as a trunk port. Set PC default gateways to their respective sub-interface IPs. Pinging from VLAN 10 to VLAN 20 now succeeds.

4. Set up static routes on the router so that a new network segment (e.g., a 'Creators' VLAN40 on a third switch) can communicate with all existing VLANs, and test connectivity with ping.

Answer & Configuration:

To connect a new network segment ('Creators' VLAN 40 on subnet 192.168.40.0/24), configure a static route on the router routing table:

```
Router(config)# ip route 192.168.40.0 255.255.255.0 192.168.10.254
```

Test connectivity by running `ping 192.168.40.10` from a PC in VLAN 10 to confirm bidirectional routing.

5. Design a mini network in Packet Tracer that mimics a Zomato-style office: create VLANsfor 'Support', 'Delivery', and 'Management', configure trunking and static routing, and document your topology with a screenshot and a brief explanation of how devices in different VLANs communicate.

Answer & Documentation:

Office Topology Design:

- **VLAN 100 ('Support'):** Subnet 192.168.100.0/24 (Customer service agents).
- **VLAN 200 ('Delivery'):** Subnet 192.168.200.0/24 (Fleet dispatch).
- **VLAN 300 ('Management'):** Subnet 192.168.300.0/24 (Administrators).

Communication Explanation:
Devices within the same department communicate locally via switch access ports. When a device in 'Support' (VLAN 100) communicates with 'Delivery' (VLAN 200), traffic is sent to the switch trunk port, received by the router's 802.1Q sub-interface (`GigabitEthernet 0/0.100`), routed across the router to sub-interface `GigabitEthernet 0/0.200`, and forwarded down to VLAN 200.